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Module 4: Software Design
Session 4: Architecture, UML and Workflow Modelling

Software Developer Level 4

What you will be able to do

By the end of this session you will be able to:

1. Analyse how layering, separation of concerns, cohesion and coupling affect maintainability.
2. Construct UML class and sequence diagrams that communicate a system's structure and behaviour.
3. Analyse a workflow with process data to find a bottleneck and justify an improvement.

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Start question from your Prepare

CHAT

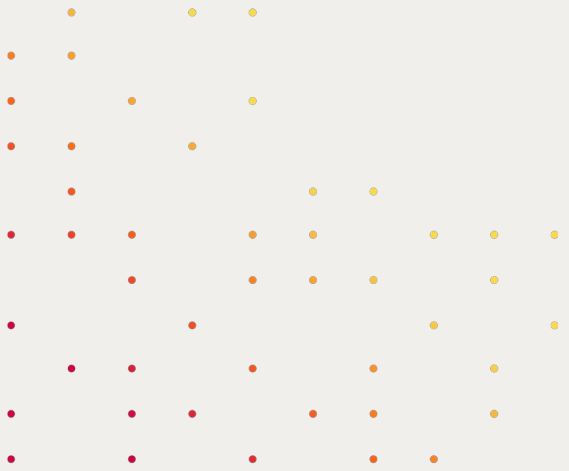
What would a contractor need from our design that a tenant would not?

One answer each in the chat. Keep the list: you will check your class diagram against it.

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Layers, cohesion and coupling

How you arrange the code decides how cheaply the portal can change.



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What Kestrel Moor needs from the structure

PRIYA NAIR, IT AND DIGITAL LEAD

> **We are four developers, so whatever we design has to be easy to change.**

Case Brief, Part 5 makes it testable:

MNT-1: add a notification channel without touching repair logic.

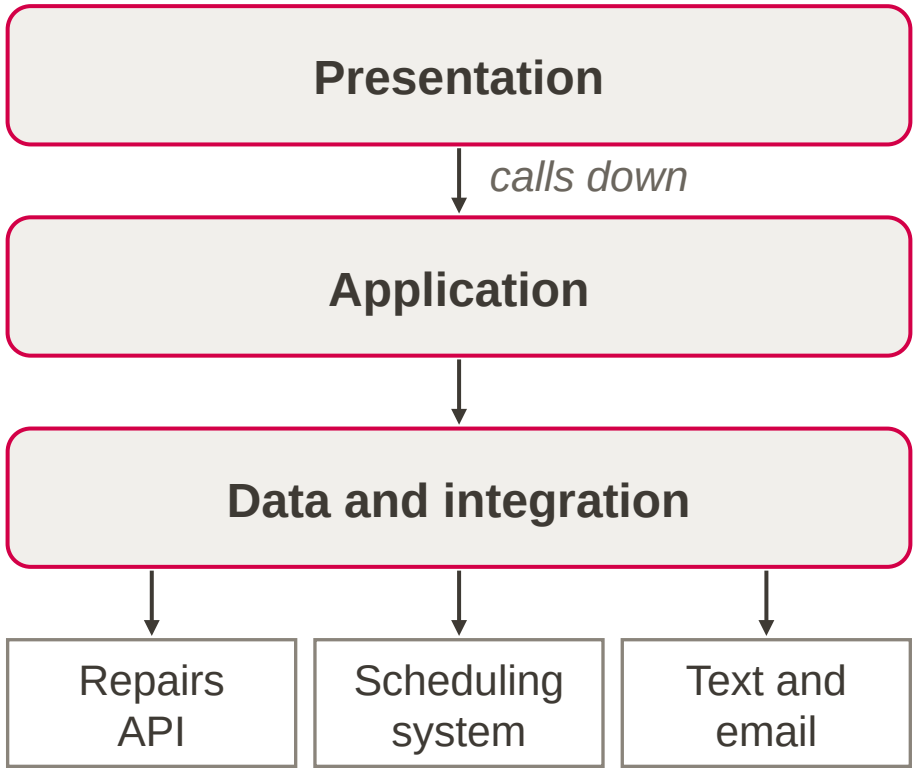
MNT-2: swap a contractor or scheduling system by changing one part.

MNT-3: keep the priority rules in one place.

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Layered architecture

Separation of concerns: each layer does one kind of job, and calls go down only.



Screens, pages and API endpoints. Show data and collect input; no rules, no SQL.

Rules, workflows and decisions. Knows what should happen, not how data is stored or shown.

Entity classes, repositories and gateways: hold and save data, and talk to other systems.

Outside systems Kestrel Moor must use (Case Brief, Part 1). Only this layer talks to them.

Cohesion and coupling

Cohesion: aim high

How closely the things inside one class serve one purpose.

High: an ArrearsRules class that only decides who is in arrears.

Low: a Helpers class that formats dates, sends texts and checks passwords.

Test: can you describe the class in one sentence without "and"?

Coupling: aim loose

How much one part depends on the inside details of another.

Loose: the rent page asks a service for the balance.

Tight: the rent page reads the payments table itself.

Test: if you change this class, how many others must change?

Worked example: a tangled rent page

Rent is outside the portal's scope: copy the method, not the classes.

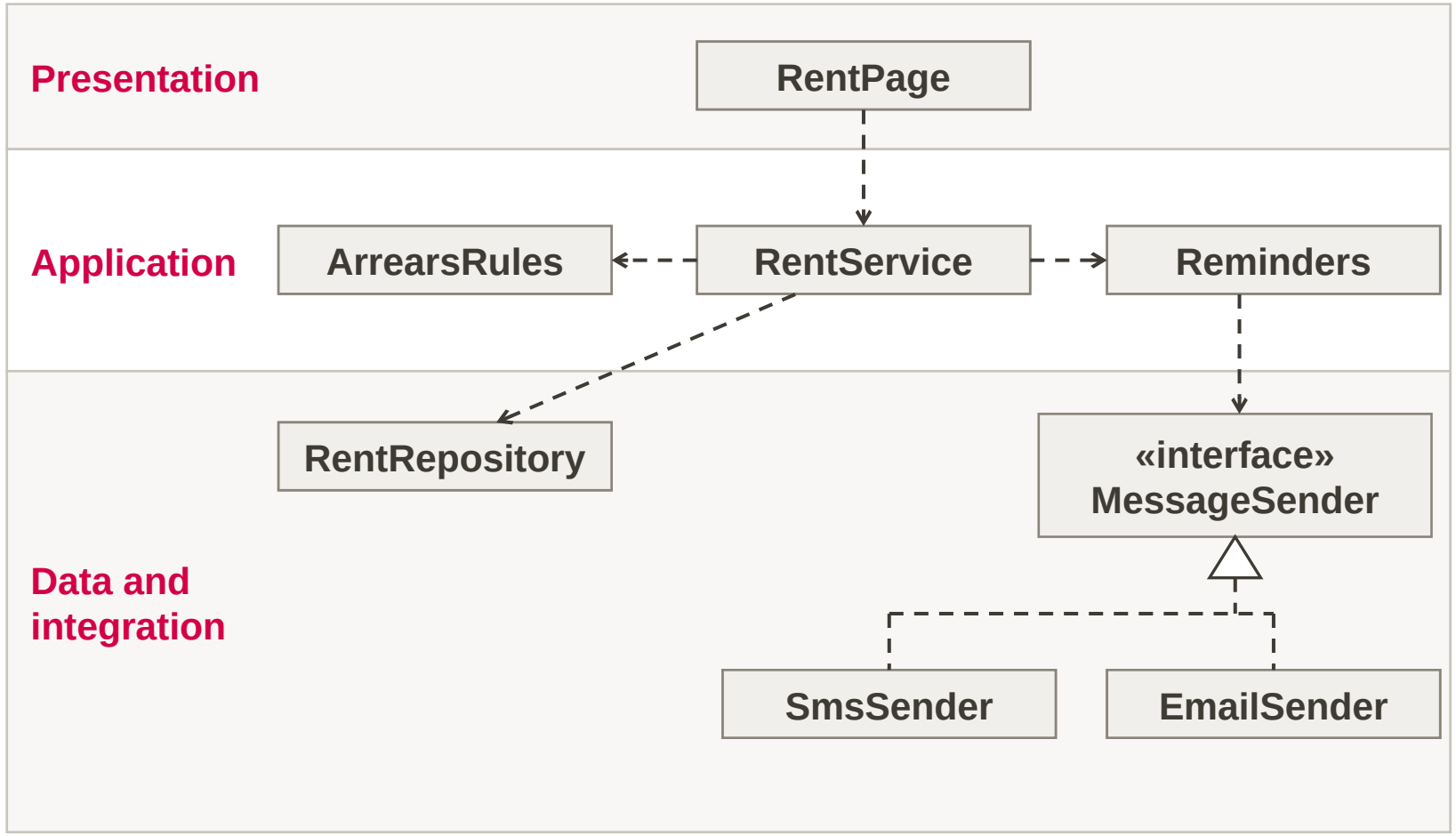
RENTPAGE (PYTHON-STYLE SKETCH)

```
    class RentPage:
      def show(self, tenant_id):
(1)     rows = db.query("SELECT amount"
          " FROM rent_txn WHERE tenant=?")
        total = sum(r.amount for r in rows)
(2)     if total > 500:
(3)       sms.send(tenant_id, "Overdue")
        self.render(total)
```

1. The page writes SQL. Rename a column and the screen breaks.
2. A policy rule hides in a screen. Change the limit and you hunt for every copy.
3. The page knows the text provider. Adding email means editing the page.

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Worked example: the same feature, layered



What a change touches

Arrears limit changes: ArrearsRules only.

WhatsApp added: one new class that implements MessageSender.

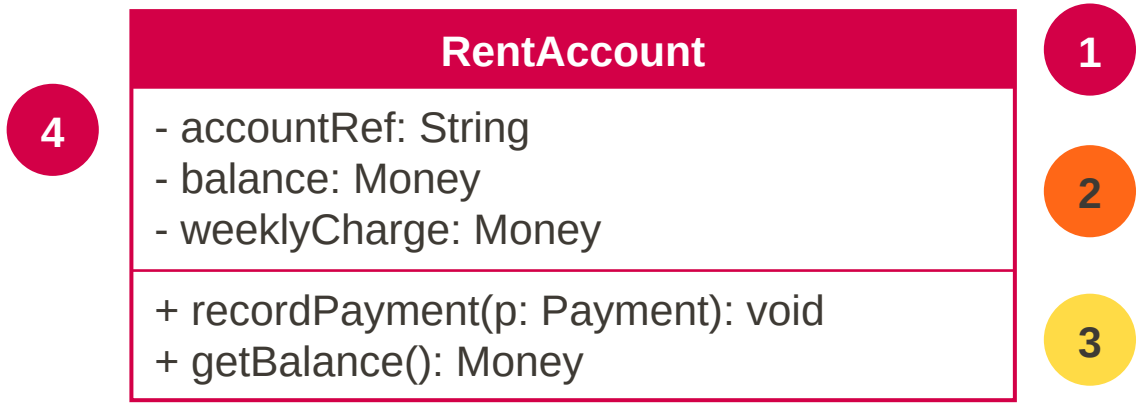
Database moves: RentRepository only.

Class diagrams

A precise way to show the parts of a system and how they relate.

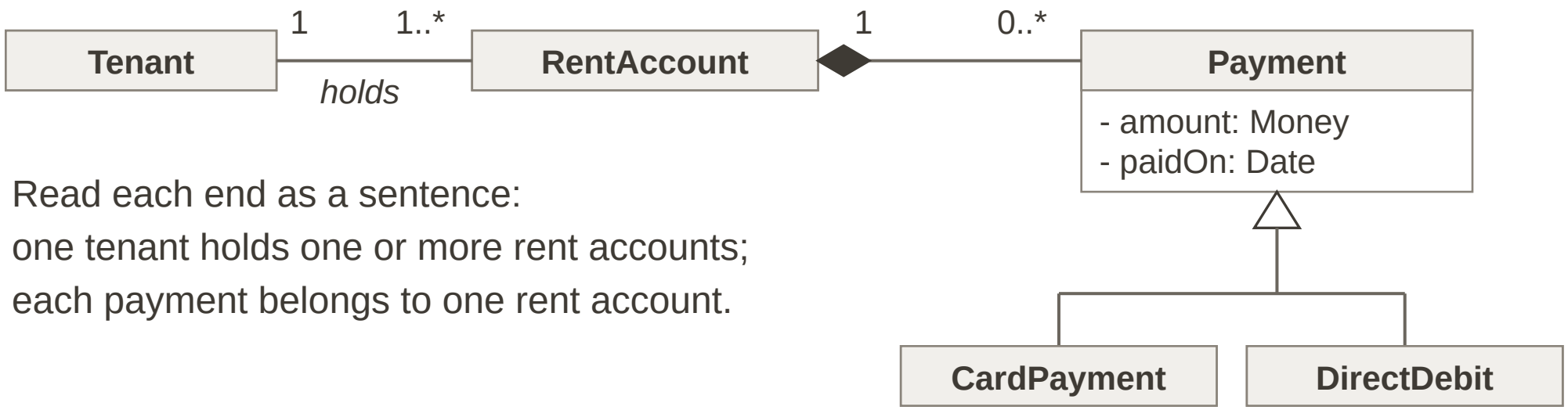
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Reading a class



1. Name: a singular noun from the domain, in UpperCamelCase.
2. Attributes: what each object knows, written name: Type.
3. Operations: what it does with its own data. Policy rules live in one rules class.
4. Visibility: - private, + public, # protected. Keep data private.

Relationships and multiplicity



Read each end as a sentence:
one tenant holds one or more rent accounts;
each payment belongs to one rent account.



Association: is linked to

Composition: owns; parts go with it

Aggregation: groups parts that live on

Generalisation: is a kind of

Realisation: implements an interface

Reading multiplicity

HOW TO READ IT

> **Start at one class and cross the line: the number at the far end says how many it has.**

Tenant 1 to 1..* RentAccount: each tenant holds one or more accounts; each account has one tenant.

1 exactly one 0..1 none or one 0..* any number 1..* at least one

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Activity 1: Model the classes

Pairs in breakout rooms

Turn your use case into a layered class diagram.

1. On your own (2 minutes): sort the nouns in your use case into data, screens and rules.
2. Together, choose six classes for the repair journey, with at least one in each layer.
3. Place each class in its layer: Presentation, Application, or Data and integration.
4. Draw the associations between your data classes, with a multiplicity at both ends.
5. Add a note: which of MNT-1 to MNT-3 your structure meets, through which class, and how.

Extension: see the Activities document

TIME
24 minutes working time

GROUPING
Pairs in breakout rooms

YOU NEED
Starter file, page "Class diagram"
Your concept map and use case (or Supporting Material A)
Supporting Material B and C

OUTPUT AND SUCCESS
Six classes in layers, with multiplicities and a maintainability note.

Debrief: Which class changes if Kestrel Moor adds WhatsApp, and what else changes?

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Break

20 minutes

Save your draw.io file to your device before you go. After the break: sequence diagrams, and which diagram suits which reader.

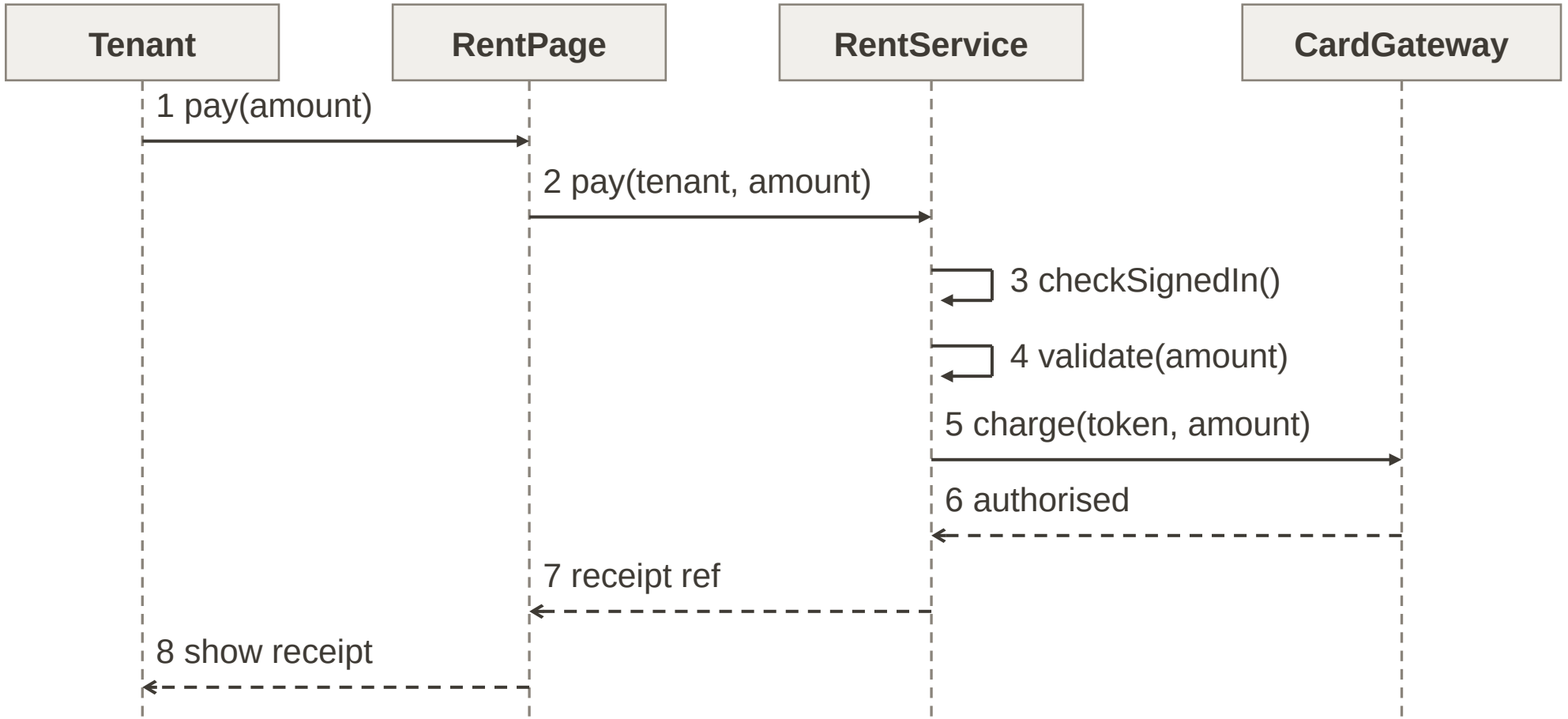
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Sequence diagrams and their readers

How to show behaviour step by step, and which diagram suits which reader.

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Worked example: paying rent online



Same system, different readers

A contact centre manager asks: who does what, in what order, and where does work wait?

Show them a swimlane or task flow.

A developer asks: which object calls which, with what data?

Show them a sequence or class diagram.

Choose the diagram from the reader's question.



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Which diagram for whom

Reader	Their question	Show them
Developers and testers	Which class does what, and who calls whom?	Class and sequence diagrams
Contact centre staff	Who does what, in what order, and where does it wait?	Task flow or swimlane
Head of Repairs	Where do we miss our targets?	Swimlane with stage times
Tenants and the Tenant Panel	What will I see and tap?	Wireframes or a clickable prototype
Contractors	What do I get for each job?	Wireframe of the job view

Activity 2: Sequence a repair report

Pairs in breakout rooms

Show how the portal handles a leak reported online.

1. Name the blank lifelines on page "Sequence" from your class diagram; add or remove any.
2. Show the messages that meet requirements R1 and R2, in an order you can justify.
3. Draw calls as solid arrows and replies as dashed arrows, and number every message.
4. Beside the diagram, write one line: why your emergency check sits where it does.

Extension: see the Activities document

TIME
18 minutes working time

GROUPING
Pairs in breakout rooms

YOU NEED
Starter file, page "Sequence"
Your class diagram
Activities: requirements R1 and R2
Supporting Material D

OUTPUT AND SUCCESS
A numbered sequence diagram a developer could build from.

Debrief: Where did you put the emergency check, and why there?

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Reading process data

Use the data you already have to find where work waits.

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Worked example: library book returns

Made-up data, not repair data. Target: every book checked in within 24 hours of return.

Book (A)	Returned (B)	Route (C)	Return to check-in, h (D)	Check-in to shelf, h (E)
LB-201	Mon 10:15	Desk	0.3	5.0
LB-202	Tue 18:30	Drop box	14.0	4.5
LB-203	Wed 11:00	Desk	0.1	6.0
LB-204	Fri 19:45	Drop box	62.0	5.5
LB-205	Thu 20:10	Drop box	13.0	4.0

=COUNTIFS(C2:C6,"Drop box",D2:D6,"<=24") returns 2: 2 of 3 drop-box books met it.

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Count the misses, not just the mean

2 of 3

drop-box books checked in within 24 hours

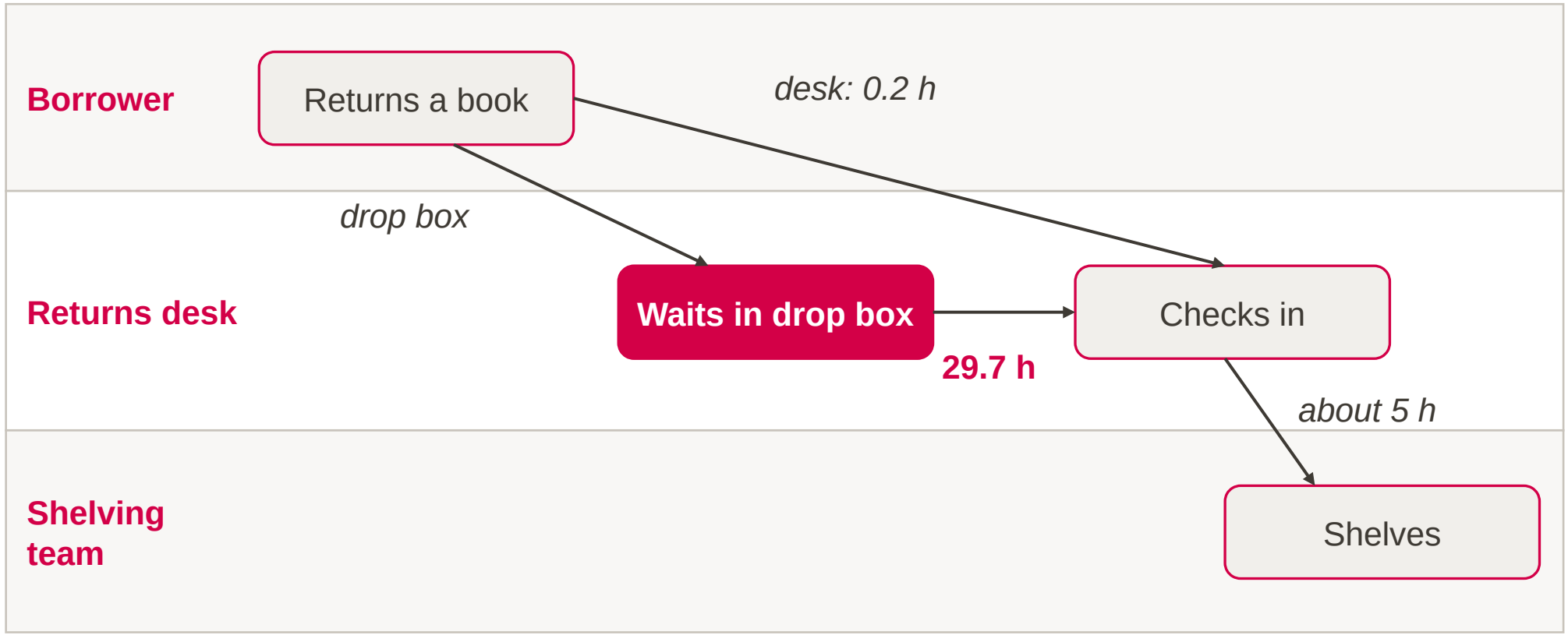
Mean 29.7 hours, over 24, though most met it. Median 14.0 hours.

All five books: mean 17.9 hours, under 24, yet one book in five missed.

For a target each item must meet, count the items within it, then give the mean and the slowest rows.

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Worked example: the returns path as a swimlane



Bottleneck: the drop-box wait. 1 of 3 missed 24 h; mean 29.7 h, median 14.0 h.

Your remit, your data

INITIATIVE AND RESPONSIBILITY

> **If the data in your system shows a problem in your area, own the next step: evidence, a fix and a measure.**

Be resourceful: a filter, COUNTIFS and AVERAGEIF on the extract you have beat waiting for a new report.

Stay in your remit: take the evidence to the person who owns the process.

Activity 3: Find the bottleneck

Pairs in breakout rooms

Use the data to find where reports wait, and fix it.

1. On your own (2 minutes): read Supporting Material G; choose the columns and target to test.
2. For each channel, work out the mean hours to triage (AVERAGEIF) and how many were triaged within 24 hours (COUNTIFS); fill Table 1.
3. Decide together which stage and channel the data points to; give your evidence in Table 2.
4. Propose a fix, a measure with a target and date, and the next step you would take yourself.

Extension: see the Activities document

TIME
22 minutes working time

GROUPING
Pairs in breakout rooms

YOU NEED
Your own copy of Repair Requests.xlsx
Activities: Tables 1 and 2
Supporting Material F to H

OUTPUT AND SUCCESS
Tables 1 and 2: where reports wait, the evidence, a fix, a measure and your next step.

Debrief: What did the count within 24 hours show that the mean alone would have hidden?

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Check your understanding

CHECK

1. Which diagram suits a contact centre manager better than a developer?

A A sequence diagram of the services
B A class diagram of the repair request data
C A swimlane of who does what, in order
D A list of the repairs API endpoints

2. A booking page queries the scheduling system's tables. What is the main risk?

A The page loses cohesion, so it loads more slowly
B The scheduling data would have to move into the page layer
C The page can no longer be tested by the testers
D Replacing the scheduling system means editing the page

3. Contractor (0..1) is linked to Operative (1..*). What does that mean?

A Each contractor has one or more operatives; each operative has no contractor or one
B Each contractor has no operatives or one; each operative has one or more contractors
C Each contractor may have no operatives; each operative has exactly one contractor
D Each contractor has one or more operatives; an operative can have many contractors

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Skills covered today

KSB and official wording	In plain English	Where we covered it today
Part of K4: how best to communicate using the different communication methods	Choosing the diagram that suits the audience	Which diagram for whom; live check
Part of K11: software designs	Reading and drawing UML class and sequence diagrams	Activities 1 and 2; live check
S8: create simple software designs to effectively communicate understanding of the program	Modelling the portal so others can understand it	Activities 1 and 2
B6: Shows initiative and takes responsibility for solving problems within their own remit, being resourceful when faced with a problem to solve.	Using the request data to find and fix a process problem	Activity 3

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Recap

1. Layers separate jobs; loose coupling keeps each change in one place.
2. A class diagram shows structure: classes, layers and multiplicities.
3. A sequence diagram shows behaviour: who calls whom, in order, with checks.
4. Choose the diagram from the reader's question.
5. For a target every item must meet, count the items within it; give the mean beside it.

Before next time

Do the Consolidate Quiz for this session (about 10 minutes; answers at the end).

Copy the key points of your Reflective Log entry (next slide) into your OneFile timesheet reflection.

Read the Session 5 Prepare: Secure by Design and Reusable Patterns - Prepare.docx

Optional: the Session 4 stretch deck.

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Reflective Log: Session 4 entry

Open Reflective Log.docx and complete the Session 4 entry now:

The bottleneck you found, the evidence for it, your improvement and the next step you would take yourself.

After the session, copy the key points into your OneFile timesheet reflection.

Keep workplace examples general: no personal or customer data, credentials or confidential material, and no employer data in AI tools your employer has not approved.

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